

Astrolabe

Calibrated reconstruction of a Parkinson’s motor diary
from bilateral wrist accelerometry

Mohammad Sadegh Salehi

26 July 2026 · astrolabe-flame.vercel.app · [source](#)

Abstract

Parkinson’s symptoms fluctuate hour to hour, but the paper diary intended to capture that is abandoned within days: in the cohort studied here only 61.6% of possible hours carry an entry, *with* research staff supporting the process. We ask whether wrist accelerometry can fill the gaps, and — more importantly — whether a system can know when it cannot.

We report two results of opposite sign. Tremor presence is recoverable across people (held-out AUC = 0.697 ± 0.075). The seven-point subjective motor scale is not: nine feature sets over five participant-level folds all fail to beat a constant predictor, and the failure mode is a sign flip in the feature–state relationship between individuals rather than insufficient capacity. We therefore do not predict it.

The contribution is the uncertainty machinery around both. Credible intervals are widened on held-out participants until empirical coverage matches the nominal level (0.903 against a 0.90 target), abstention is tuned to a fixed *error budget* rather than a fixed rate, and selective prediction is measured rather than asserted: accuracy rises from 0.713 to 0.825 as the answered fraction falls to a quarter. Removing one wrist forces abstention from 12.4% to 77.3% to hold the same budget — degradation the system discovers rather than is told.

1 Data

COPS (Continuous Observation of Parkinsonian Symptoms; Nesser et al., CC-BY 4.0) provides 66 participants wearing GENEActiv accelerometers on both wrists at 100 Hz, paired with hourly self-reported symptom diaries over roughly a week each. After excluding participants without usable paired data, 6,530 hours across 65 participants remain.

The alignment convention. A diary row stamped T describes the window $[T - 1, T)$. This is stated once in the study documentation and is easy to invert; doing so shifts every label by one hour and silently degrades every downstream result. `io_cops.py` re-derives the window from the filename and raises on mismatch, so a misaligned join cannot be constructed.

Cohort caveat. 46 of 66 participants have deep brain stimulation. This is an advanced, device-treated population; both the movement signature and the fluctuation pattern differ from newly diagnosed patients, and nothing here should be assumed to transfer unchanged.

2 Features

Each wrist signal is band-passed into the two bands the COPS authors themselves use: a movement band 0.1–3 Hz and a tremor band 4–8 Hz. Band power, spectral ratios, and bilateral asymmetries are computed over 10-minute windows, giving 122 features.

Butterworth filters of order 8 are applied as second-order sections; the transfer-function form is numerically unstable at this order and returns NaN silently, which is a failure that propagates invisibly to the model rather than crashing.

3 Method

3.1 Ordinal emissions

The kinesia scale is ordinal with $K = 7$ levels. Following Frank and Hall we decompose it into $K - 1$ cumulative binary problems, fit a gradient-boosted classifier to each, and difference the results:

$$\Pr(z = k \mid x) = \Pr(z > k - 1 \mid x) - \Pr(z > k \mid x), \quad (1)$$

enforcing monotonicity of the cumulative estimates by a running minimum before differencing, so no negative mass can appear.

3.2 Dividing out the prior

A discriminative classifier returns $p(z \mid x)$, but the state-space model below requires an emission likelihood $p(x \mid z)$. By Bayes' rule, up to a factor constant in z ,

$$p(x \mid z) \propto \frac{p(z \mid x)}{p(z)}. \quad (2)$$

Omitting the division counts the class prior twice — once in the classifier and once in the chain — which biases the posterior toward the majority state in a way that looks like smoothing working well.

3.3 Temporal model

Motor state is persistent, so we place a hidden Markov model over the day and recover the posterior by forward-backward in log space. The transition matrix is built as a convex combination rather than a matrix root,

$$T = (1 - s)I + sB, \quad (3)$$

with B the empirical hour-to-hour transition matrix and switch rate $s = 0.02$. A matrix root of B can contain negative entries and is not a valid stochastic matrix at fine time steps.

Likelihoods are tempered before smoothing,

$$\tilde{\ell}(x \mid z) = \frac{1}{\tau} \log p(x \mid z), \quad \tau = 25, \quad (4)$$

selected on a held-out fold. Without tempering the chain is over-confident and the entire day collapses onto a single state.

3.4 Calibrated credible intervals

For posterior γ at a step we take the shortest contiguous interval carrying mass m ,

$$[\ell, u] = \arg \min_{u-\ell} \left\{ \sum_{k=\ell}^u \gamma_k \geq m \right\}. \quad (5)$$

Crucially m is *not* set to the nominal 0.90. Requesting 90% of posterior mass does not deliver 90% empirical coverage, and asserting otherwise is precisely the unfalsifiable claim this system exists to avoid. Instead m is swept on participants held out of *both* the emission fit and the test fold until achieved coverage reaches the target:

$$m^* = \min \left\{ m : \frac{1}{n} \sum_i \mathbf{1}[z_i \in [\ell_i(m), u_i(m)]] \geq 0.90 \right\}. \quad (6)$$

This yields $m^* = 0.72$ and achieved coverage 0.903.

Configuration	Mass m^*	Coverage	Abstains	MAE answered
Both wrists	0.72	0.903	12.4%	0.226
Left wrist dropped	0.89	0.904	77.3%	0.218

Table 1: Calibration is fitted *per sensor configuration* on 6,838 hours from 11 held-out participants. Both hold the same error budget; the answer rate is what moves.

A bug worth reporting. Interval mass was initially fitted on the raw emission posterior and applied to the smoothed HMM output. The two distributions have entirely different sharpness, and coverage landed at 0.535 against a 0.90 target — which the interface would have displayed as “90%”. Calibration and inference now share one code path (`infer_posterior`) so the two cannot diverge.

3.5 Abstention as a fixed error budget

A step is declined when the posterior is uninformative:

$$\text{abstain} \iff \max_k \gamma_k < \pi \quad \vee \quad (u - \ell + 1) \geq w. \quad (7)$$

The threshold π is *not* tuned to a target abstention rate. Doing so guarantees that every sensor configuration declines equally often, so a degraded configuration cannot exhibit degradation — the threshold simply absorbs it. We observed exactly this: with one wrist blanked, abstention *fell* from 12% to 46% in the wrong direction, because gradient-boosted trees route missing features down a default branch and return a *sharper* posterior on strictly less evidence.

Instead π holds a fixed risk on answered steps. Writing $\mathcal{A}(\pi)$ for the answered set and $\hat{z}_i = \gamma_i^\top(0, \dots, 6)$,

$$\pi^* = \min \left\{ \pi : \frac{1}{|\mathcal{A}(\pi)|} \sum_{i \in \mathcal{A}(\pi)} |\hat{z}_i - z_i| \leq \beta \right\}, \quad (8)$$

with budget β set by the reference configuration. A worse configuration can only meet β by answering less, so the abstention rate becomes a consequence of measured accuracy.

4 Results

4.1 Tremor generalises; subjective state does not

Target	Metric	Held-out
Tremor present (TremorScore > 0)	AUC	0.697 ± 0.075
Kinesia OFF/ON, identical folds	AUC	0.567
Kinesia, 7-state reconstruction	ordinal MAE	0.684
constant baseline (always “Good”)	ordinal MAE	0.594
per-participant oracle	ordinal MAE	0.380

Table 2: Tremor is a 4–8 Hz mechanical oscillation the accelerometer measures directly. How well an hour *felt* is a judgement the sensor has no access to.

4.2 Why the subjective scale fails

The cumulative classifiers reveal the mechanism. Held-out AUCs for $\Pr(z > k)$ are 0.428, 0.530, 0.587, 0.324, 0.455 — three of five *below chance*. A classifier at 0.5 has no signal; one at 0.324 has signal pointing the wrong way. The feature-to-state mapping inverts between individuals, and no amount of regularisation repairs a sign that flips.

Personalisation does not rescue it. Given identical hours and the *same* calibration labels, a participant’s own constant scores ordinal MAE 0.400 while the global model plus a per-participant offset scores 0.474. The offset was learning the person’s mean; the global mapping it shifted added nothing.

4.3 The model was over-featured

Feature set	Held-out tremor AUC
All 122 features	0.672
minus medication timing	0.688
minus medication timing and environment	0.702
50 band-power features only	0.722

Table 3: Medication-timing features alone score AUC = 0.433 — *below* chance. Who takes which drug on what schedule is close to a participant fingerprint and cannot transfer. The shipped detector reserves a fifth of the training participants to fit its calibrator, costing about 0.025 AUC and yielding the 0.697 quoted throughout.

4.4 Selective prediction

Abstention earns its place only if the declined hours are the ones that would have been wrong.

Answered fraction	n	Accuracy	Balanced acc.
1.00	6,291	0.713	0.610
0.75	4,916	0.758	0.649
0.50	3,312	0.785	0.687
0.25	1,624	0.825	0.708

Table 4: Balanced accuracy is reported because at 30.5% prevalence raw accuracy can be bought by predicting the majority class.

4.5 The caveat that must be stated

Pooled out-of-fold AUC is 0.681, but the median *within-participant* AUC is 0.550, above chance on 25 of 36 scorable participants. Much of the pooled figure separates tremor-dominant people from everyone else rather than a tremulous hour from a calm one *in the same person* — and within-person discrimination is what a diary requires. Both figures ship in the output contract.

5 What the system will not claim

The seven-point motor scale is never inferred. It is displayed as reported by the patient, or as an explicit abstention. On the demonstration participant-day, the model declines all 114 steps, its peak posterior never exceeding 0.549 against a threshold of 0.58 fixed on other participants. That is the system behaving correctly.

No causal claim is made. Trajectory shape around recorded medication times is temporal alignment, not treatment response; establishing response requires a controlled protocol. This is not a medical device.

6 Reproducibility

Every figure above is regenerated by a named script.

Script	Establishes
<code>baselines.py</code>	the constant and oracle baselines
<code>diary_completion.py</code>	the 61.6% completion rate
<code>diagnose_deep.py</code>	tremor versus kinesia on identical folds
<code>diagnose_featureset.py</code>	the nine-feature-set sweep
<code>diagnose_fair.py</code>	offset versus constant given equal labels
<code>train_tremor.py</code>	the shipped detector and selective curve
<code>make_bundles.py</code>	calibration, abstention, and the output contract

Data: Nesser, T. et al., *COPS — Continuous Observation of Parkinsonian Symptoms*, OSF (2025), 10.17605/OSF.IO/5XVWN, CC-BY 4.0. Code: MIT. Not a medical device; no diagnostic, dosing or treatment claim is made.